



Punjab University College
of
Information Technology
“PUCIT”

Chapter 16 – Intellectual Property Rights, Ethics, Health, Access, and the Environment

GE-161 Introduction to Information and Communication Technologies

Prof. Dr. Syed Waqar ul Qounain
Department of Information Technology
University of the Punjab, Lahore

Overview

- This chapter covers:
 - Various types of intellectual property rights
 - A discussion of ethics, including ethical use of copyrighted material, resources and information, unethical use of digital manipulation, and ethical business practices
 - The impact of computers on our physical and emotional health
 - Issues related to the access of technology
 - The impact of computers on our environment
 - A look at legislation related to these issues

Intellectual Property Rights

- Intellectual property rights: Rights to which creators of original creative works are entitled
- Copyrights: Form of protection available to the creator of original artistic or literary works
 - Last until 70 years after creator's death
 - For corporate copyrights or anonymous works, last 95 years from date of publication or 120 years from date of creation, whichever is shorter
 - Can be registered with U.S. Copyright Office

FIGURE 16-1
Copyright statements.
Statements such as these are often included on books, Web sites, and other original copyrighted works.

© 2010 Course Technology, Cengage Learning

ALL RIGHTS RESERVED. No part of this work covered by the copyright herein may be reproduced, transmitted, stored or used in any form or by any means graphic, electronic, or mechanical, including but not limited to photocopying, recording, scanning, digitizing, taping, Web distribution, information networks, or information storage and retrieval systems, except as permitted under Section 107 or 108 of the 1976 United States Copyright Act, without the prior written permission of the publisher.

BOOK COPYRIGHT NOTICES

© Red Lobster. All rights reserved.

WEB SITE COPYRIGHT NOTICES

 Watermarkly

Intellectual Property Rights

- To protect rights, can use:
 - Digital watermarks: Subtle alteration of digital content that is not noticeable but that can identify the copyright holder
 - Digital rights management (DRM) software: Controls use of the copyrighted work
 - Can limit who can view, print or copy a document
 - Can control use of downloaded content (number of devices a file can be copied to, expiration of VOD movie, etc.)

Intellectual Property Rights

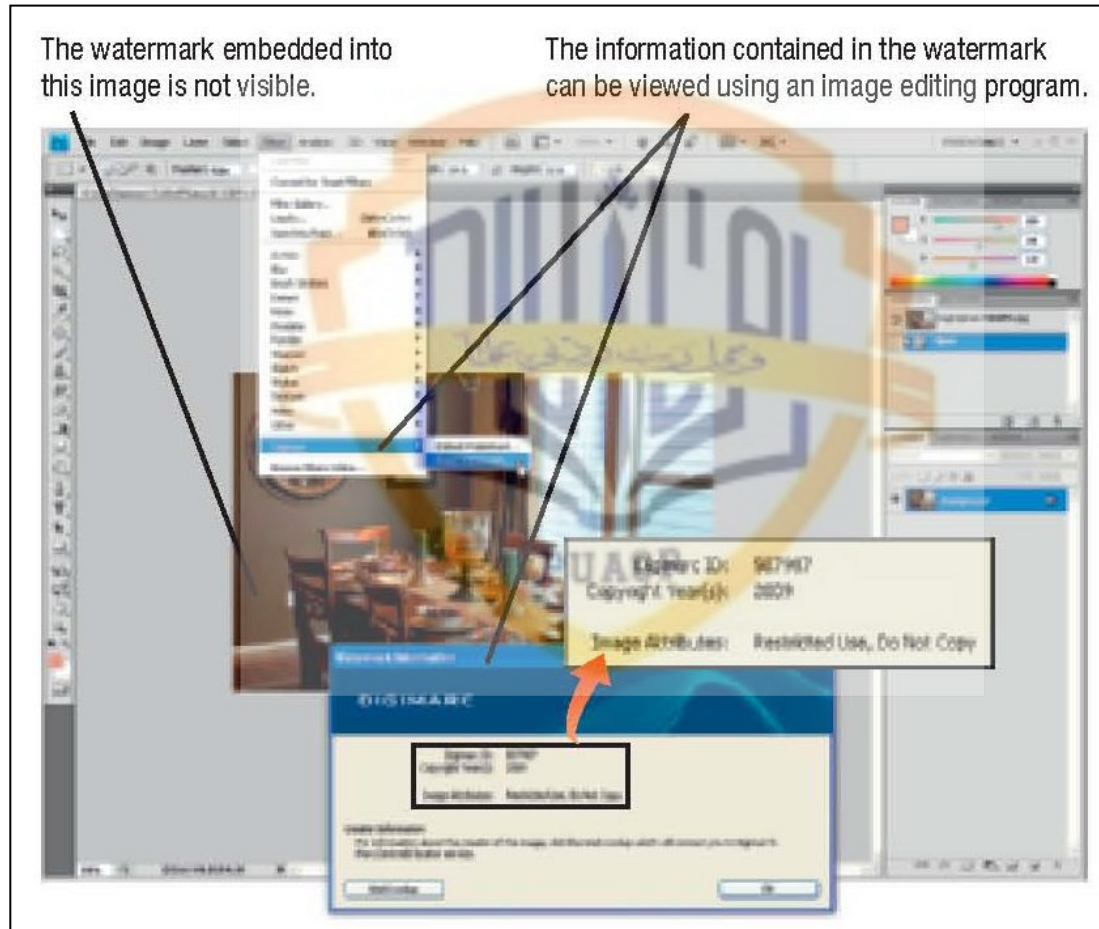


FIGURE 16-2
Digital watermarks.

Intellectual Property Rights

- Trademark: A word, phrase, symbol, or design that identifies goods or services
 - Includes protection for domain names (cybersquatting and typosquatting)
 - Domain name disputes can be brought to the World Intellectual Property Organization (WIPO)



Intellectual Property Rights

- Patents : Protect inventions
 - Lasts for 20 years
 - Can also protect a business practice or procedure
- PriceLine.com's name-your-own-price
- Amazon.com's one-click purchase
- Expensive to get but can be very lucrative

FIGURE 16-4
Patents. The patent shown here is for a new mobile phone.



Quick Quiz

1. Copyrights are valid for _____.
 - a. 70 years after the creation of the work
 - b. 70 years after the publication of the work
 - c. 70 years after the death of the creator
2. True or False: Logos cannot be trademarked, just the names of companies or products.
3. _____ are used to protect inventions.

Answers:

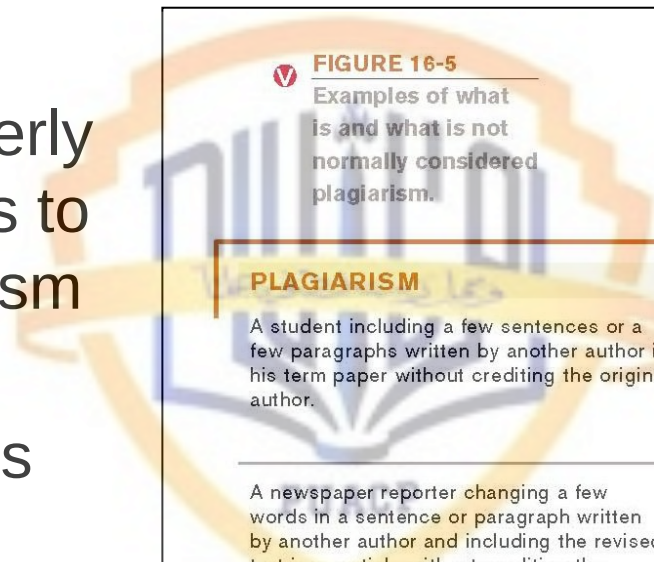
1) c; 2) False; 3) Patents

Ethics

- Ethics: Overall standards of moral conduct
 - Can vary with individual and religious beliefs, country, race, or culture
 - Personal ethics: Guide an individual's personal behavior
 - Business ethics: Guide a business's policies, decisions, and actions
 - Computer ethics: Concern moral conduct related to computer use
 - Individuals and businesses need to make ethical decisions every day

Ethical Use of Copyrighted Material

- Books and Web-based articles
 - Need to properly credit sources to avoid plagiarism
 - Strict consequences for plagiarism
 - Online tests for plagiarism are available and widely used by schools

 FIGURE 16-5 Examples of what is and what is not normally considered plagiarism.	
PLAGIARISM	NOT PLAGIARISM
A student including a few sentences or a few paragraphs written by another author in his term paper without crediting the original author.	A student including a few sentences or a few paragraphs written by another author in his term paper, either indenting the quotation or placing it inside quotation marks, and crediting the original author with a citation in the text or with a footnote or endnote.
A newspaper reporter changing a few words in a sentence or paragraph written by another author and including the revised text in an article without crediting the original author.	A newspaper reporter paraphrasing a few sentences or paragraphs written by another author without changing the meaning of the text, including the revised text in an article, and crediting the original author with a proper citation.
A student copying and pasting information from various online documents to create her research paper without crediting the original authors.	A student copying and pasting information from various online documents and using those quotes in her research paper either indented or enclosed in quotation marks with the proper citations for each author.
A teacher sharing a poem with a class, leading the class to believe the poem was his original work.	A teacher sharing a poem with a class, clearly identifying the poet.

Ethical Use of Copyrighted Material

Instructor submits electronic versions of student papers; the results are usually available online almost immediately.

The black text was correctly identified as being original.

The red text was correctly identified as being taken from a HowStuffWorks.com online article.

The green text was correctly identified as being taken from a Wikipedia.com definition.



FIGURE 16-6
Results of an online originality test on an essay that contained plagiarized content.

Ethical Use of Copyrighted Material

- Music
 - Debate began with Napster
 - Still concerns surrounding P2P file sharing sites
 - Copying purchased songs for personal use usually considered fair use
 - DRM controls can impact files, purchased CDs, etc.
 - Many downloads today are DRM-free MP3 format

✓ **FIGURE 16-7**
Many music stores today offer DRM-free song downloads.



Ethical Use of Copyrighted Material

- Movies
 - Movie piracy is high (illegal copies, recording pre-release movies to create DVDs, etc.)
 - Distributing bootleg copies of movies is illegal and unethical
 - DRM controls on downloaded movies can slow piracy
 - Legal prosecution for copyright violations for both music and movies is increasing

Ethical Use of Resources and Information

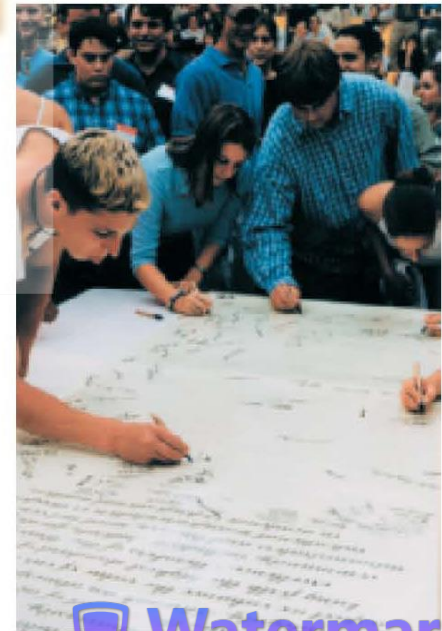
- School or company resources
 - Code of conduct: Policy that specifies allowable use of resources by students or employees
 - Students and employees should be familiar with what is considered acceptable use of resources
 - Code of ethics: Policy, typically for an industry or organization, that specifies overall moral guidelines adopted by that industry or organization

Ethical Use of Resources and Information

- Employee and customer information
 - Businesses need to decide what is ethical use of employee and customer information
 - Business schools are increasing business ethics
- Cheating and falsifying information
 - Cheating at high schools and colleges is rampant
 - Can be reduced by academic honor codes
 - Résumé padding is considered unethical by most companies

FIGURE 16-10

Academic honor codes. The honor code at the University of Denver is signed by virtually all incoming students.



Computer Hoaxes and Digital Manipulation

- Computer hoax: An inaccurate statement or story spread through the use of computers
 - Often sent via e-mail
 - Often related to viruses, health issues, political issues, etc.
 - Consider researching before passing on to others

FIGURE 16-11
Hoax-Slayer. This is one site that can be used to research possible computer hoaxes.



Computer Hoaxes and Digital Manipulation

- Digital manipulation: Digitally altering text, images, photographs, music, and other digital content
 - Some beneficial ethical uses (aging photos of runaways, altering photos of wanted criminals, etc.)
 - Use by media is more controversial
 - Difficult to tell in the future if a historical photo was manipulated



FIGURE 16-12

Digital manipulation.

The digitally manipulated photo (left) added an additional missile launching to the real photo (right) and appeared on the front page of many major newspapers.



DIGITALLY ALTERED PHOTO



ORIGINAL PHOTO

Ethical Business Practices and Decision Making

– Cultural considerations

- Ethics vary within a country as well as from country to country
- Some acts may be socially acceptable or ethical in one country, but not another
- Laws also vary from country to country
- Individuals and businesses need to consider both legal and ethical issues in global transactions
- Some business schools and corporations are including diversity and cross-cultural training

FIGURE 16-14
Cultural considerations.

In some countries, bootleg copies of music CDs and movie DVDs are sold openly, such as this DVD seen recently at a store in China.



Quick Quiz

1. An inaccurate statement or story spread through the use of computers is referred to as _____.
 - a. digital manipulation
 - b. code of ethics
 - c. computer hoax
2. True or False: Most legal experts agree that it's okay for someone who has legally obtained an audio CD to transfer those songs to a CD-R disc for personal use.
3. Software and hardware products that have been announced and advertised, but which are not yet available are considered _____.

Answers:

1) c; 2) True; 3) vaporware

Computers and Health

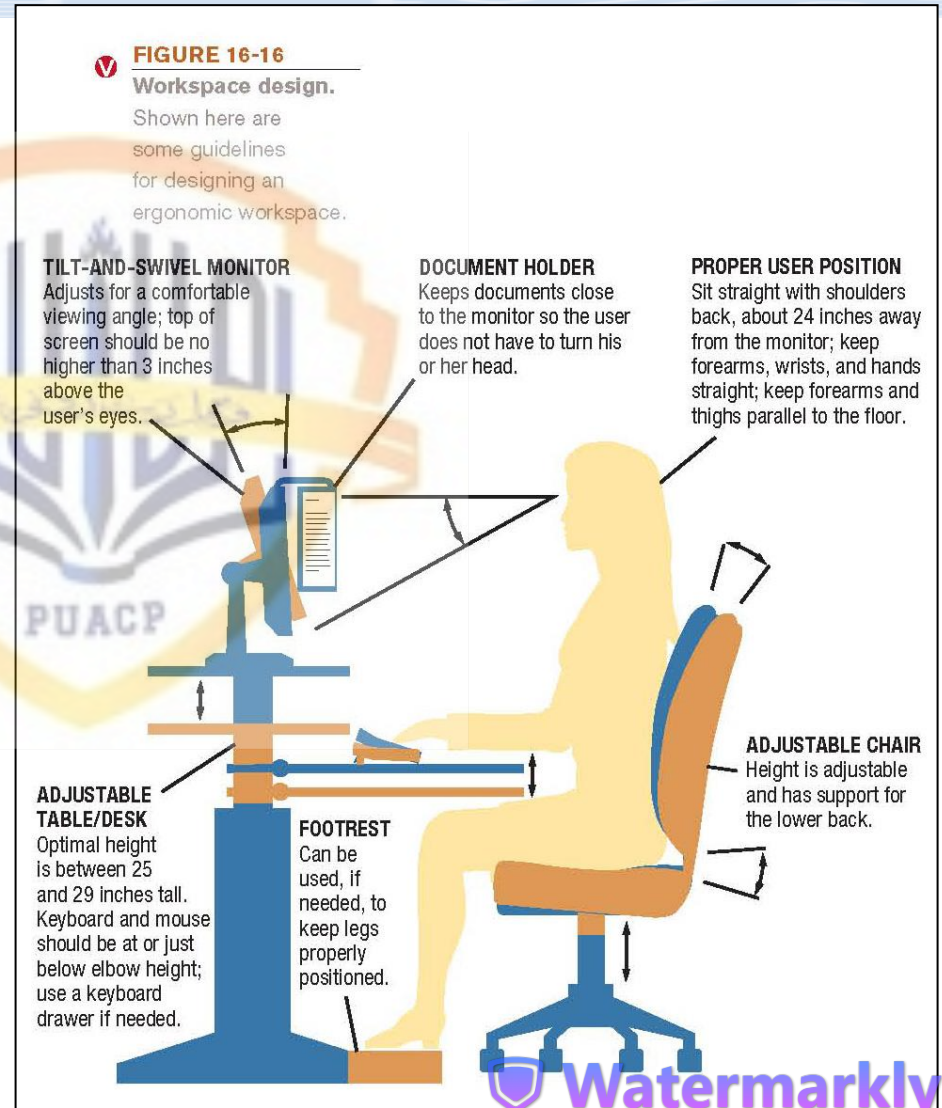
- Computer use can cause physical injuries
 - Repetitive stress injury (RSI)
 - Carpal tunnel syndrome (CTS) (keyboard use)
 - DeQuervain's tendonitis (mobile device keyboard and thumbpad use)
 - Computer vision syndrome (CVS)
 - Backaches
 - Heat from laptops
 - Hearing loss from headphones
 - Phone and texting-related car accidents
 - Possible radiation risks



FIGURE 16-15
DriveAssist. This product deactivates a mobile phone when the car is in motion.

Computers and Health

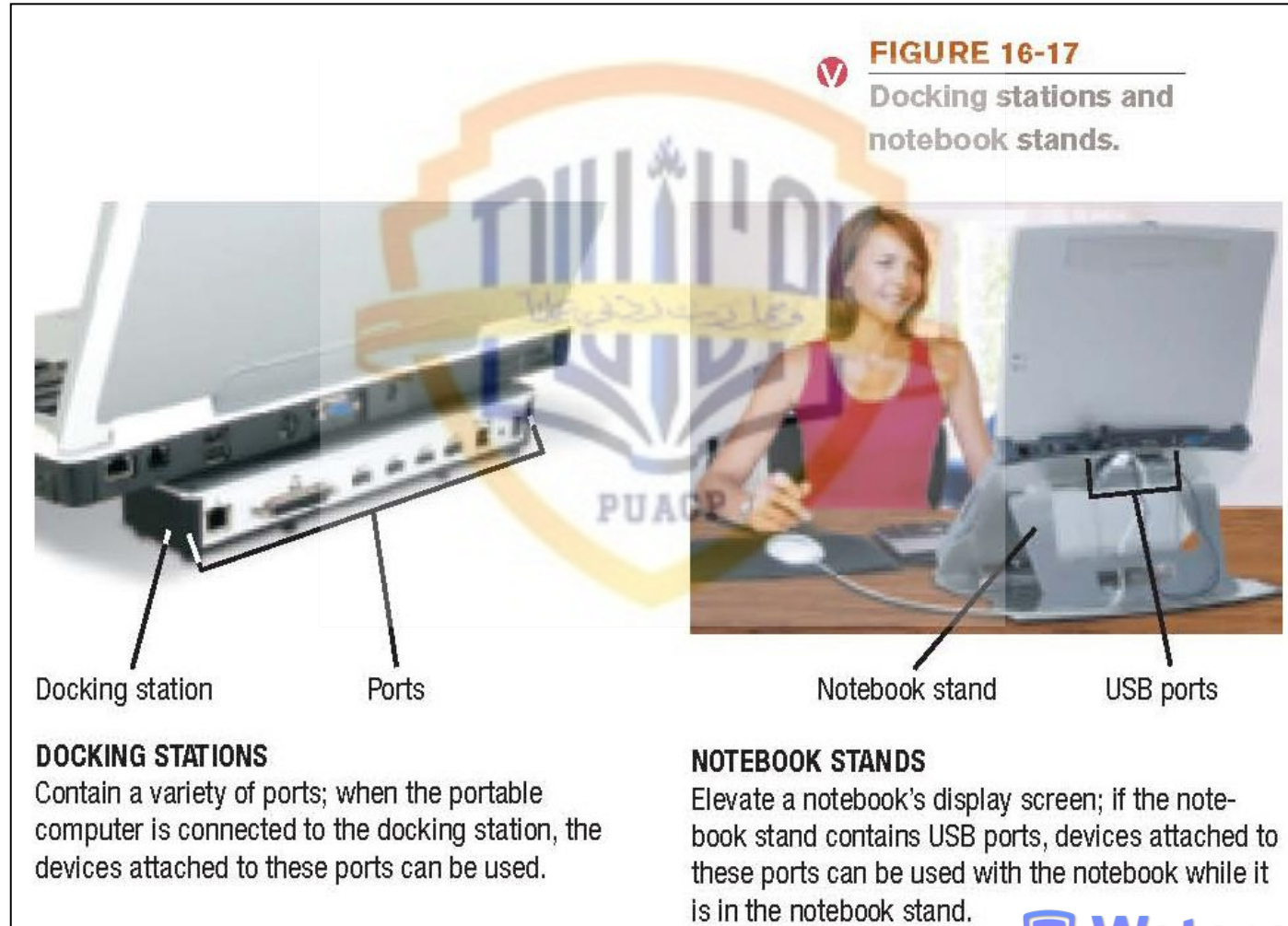
- Ergonomics: The science of fitting a work environment to the people who work there
 - Using good workspace design principles can help avoid physical problems
 - Ergonomic chairs, etc. are available



Computers and Health

- More difficult with portable computers and mobile devices, but possible to improve work environment
 - Travel mice and travel keyboards can help while on the go
 - Docking stations or notebook stands can be used with portable computer at home or in the office
 - Docking station connects the computer to permanent hardware, such as a keyboard, mouse, monitor, etc.
 - Notebook stand raises a notebook up to a better height

Workspace Design



Workspace Design

OCCASIONAL USERS

Sit with the computer on a table and position it for comfortable wrist posture. If no table is available, use a notebook cooling stand to protect your legs from the computer's heat.

Adjust the screen to a comfortable position, so you can see the screen as straight on as possible. If you have a portable notebook stand, use it to elevate the display screen for easier viewing.

Bring a travel keyboard and mouse to use with the computer, whenever possible.

When purchasing a portable computer, pay close attention to the total weight of the system (computer, power supply, additional hardware, etc.) if you will be using the computer primarily while traveling; purchase a lightweight system to avoid neck and shoulder injuries when carrying the computer from one location to another.

FULL-TIME USERS

Sit with the computer on a desk or table and position it for comfortable wrist posture if you won't be using a separate keyboard and mouse.

Elevate the computer so the screen is at the proper height, or connect the computer to a stand-alone monitor instead of using the computer's built-in display; consider using a docking station or notebook stand.

Use a separate keyboard and mouse, either attached directly to the computer or to a docking station or notebook stand.



FIGURE 16-18

Ergonomic tips for occasional and full-time portable computer users.

Ergonomic Hardware

- A variety of devices available that are designed to avoid physical problems due to the use of a computer
 - Ergonomic keyboards and trackballs
 - Document holders
 - Antiglare screens
 - Keyboard drawers
 - Wrist supports
 - Computer gloves
- Good user habits can also help avoid problems

Ergonomic Hardware

FIGURE 16-19
Ergonomic devices.



ERGONOMIC KEYBOARDS



TRACKBALLS



DOCUMENT HOLDERS



ANTIGLARE SCREENS



KEYBOARD DRAWERS



WRIST SUPPORTS



COMPUTER GLOVES

Good User Habits

FIGURE 16-20 Good user habits. These preventative measures can help avoid discomfort while working on a computer.	
CONDITION	PREVENTION
Wrist/arm/hand soreness and injury	➤ Use a light touch on the keyboard. ➤ Rest and gently stretch your fingers and arms every 15 minutes or so. ➤ Keep your wrists and arms relaxed and parallel to the floor when using the keyboard. ➤ When using a device with a small keyboard, type short messages, take frequent breaks, and use a separate keyboard whenever possible. ➤ Use an ergonomic keyboard, ergonomic mouse, computer gloves, and other ergonomic devices if you begin to notice wrist or hand soreness.
Eyestrain	➤ Cover windows or adjust lighting to eliminate glare. ➤ Concentrate on blinking your eyes more often. ➤ Rest your eyes every 15 minutes or so by focusing on an object in the distance (at least 20 feet away) for one minute and then closing your eyes for an additional minute. ➤ Make sure your monitor's brightness and contrast settings are at an appropriate level. ➤ Use a larger text size or lower screen resolution, if needed. You should be able to read what is displayed on your monitor from three times the distance at which you normally sit.

Good User Habits

Sore or stiff neck	➤ Use good posture. ➤ Place the monitor and any documents you need to refer to while using your computer directly in front of you. Use a document holder if possible. ➤ Adjust your monitor to a comfortable viewing angle with the top of the screen no higher than 3 inches above your eyes. ➤ Use a telephone headset if you spend a significant amount of time each day on the telephone.
Backache; general fatigue	➤ Use good posture and adjust your chair to support your lower back; use an ergonomic chair, if needed. ➤ Use a footrest, if needed, to keep your feet flat on the floor. ➤ Walk around or stretch briefly at least once every hour. ➤ Alternate activities frequently. ➤ When traveling with a computer, bring a lightweight notebook or netbook computer and carry only the essentials with you.
Ringing in the ears; hearing loss	➤ Turn down the volume when using headphones (you should be able to hear other people's voices). ➤ Wear over-the-ear-headphones instead of earbuds. ➤ Limit the amount of time you use headphones or earbuds. ➤ Use external speakers instead of headphones when possible.
Leg discomfort or burns	➤ Use a laptop desk or other barrier between a portable computer and your legs when using a computer on your lap.

FIGURE 16-20
Good user habits.

These preventative measures can help avoid discomfort while working on a computer.

Computers and Health

- Computer use can also be a problem for emotional health
 - Stress due to ever-changing technology
 - Many jobs require computer use now that didn't in the past
 - Available 24/7 via technology

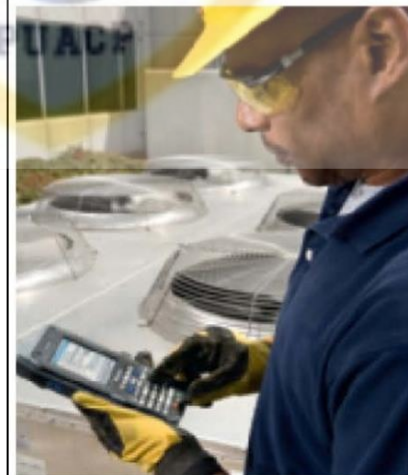
FIGURE 16-21
Computer-related jobs. Many jobs that did not require computer use in the past require computer use today.



RESTAURANT SERVERS



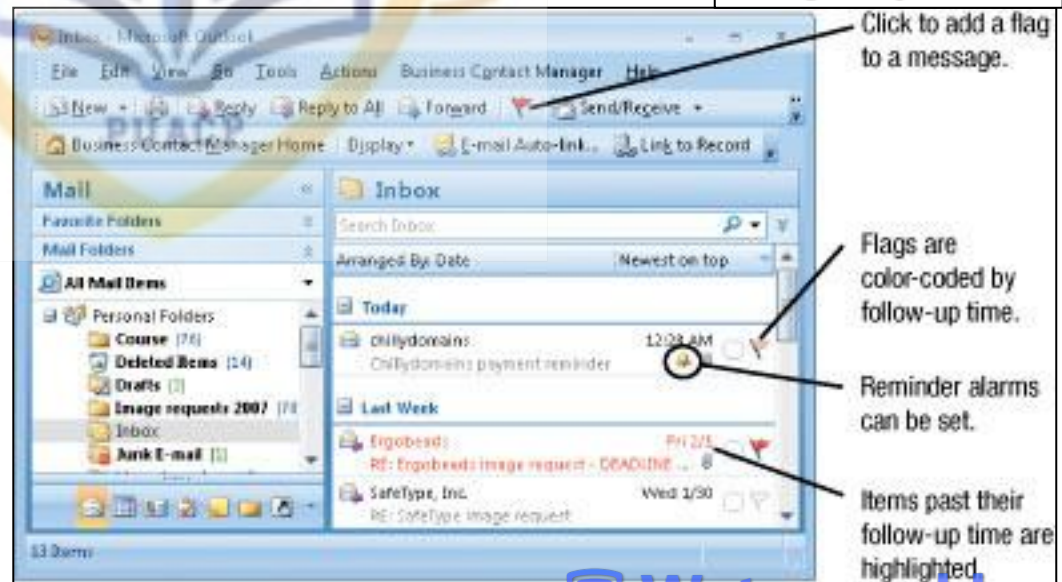
POLICE OFFICERS



Computers and Health

- Information overload
 - Good searching techniques can help
 - Managing your e-mail can help (e-mail filters, flags, checking only periodically, etc.)
- Burnout: A state of fatigue or frustration brought on by overwork
 - Reevaluate schedule, ask for help, keep healthy food and exercise routines, etc.

FIGURE 16-23
Outlook reminder flags can help you organize your Inbox.



Computers and Health

- Internet addiction: The problem of overusing, or being unable to stop using, a computer or the Internet
 - Growing problem in many countries
 - Often related to e-mail, IM, shopping, gaming, social networking, and pornography
 - Can be addicted to other types of technology (video games, etc.)
 - Can result in loss of relationships, job loss, academic failure, health problems, financial consequences, child custody, suicide, and more
 - Can be treated similar to other addictions

Computer/Internet Addiction

Do you feel preoccupied with the Internet (think about previous online activity or anticipate next online session)?

Do you feel the need to use the Internet with increasing amounts of time in order to achieve satisfaction?

Have you repeatedly made unsuccessful efforts to control, cut back, or stop Internet use?

Do you feel restless, moody, depressed, or irritable when attempting to cut down or stop Internet use?

Do you stay online longer than originally intended?

Have you jeopardized or risked the loss of a significant relationship, job, educational, or career opportunity because of the Internet?

Have you lied to family members, a therapist, or others to conceal the extent of involvement with the Internet?

Do you use the Internet as a way of escaping from problems or of relieving a dysphoric mood (e.g., feelings of helplessness, guilt, anxiety, depression)?

FIGURE 16-24
Signs of Internet addiction. You may be addicted to the Internet if you answer "yes" to at least five of these questions.

Access to Technology

- Digital divide: The gap between those who have access to technology and those who don't
- Can have digital divide within a country, as well as between countries
- U.S. digital divide: Differences in technology use by income, race, geographical area, education, broadband Internet use, etc.
 - Reducing the U.S. digital divide is important to ensure all citizens have an equal chance of being successful
 - Some people choose not to use technology

The U.S. Digital Divide

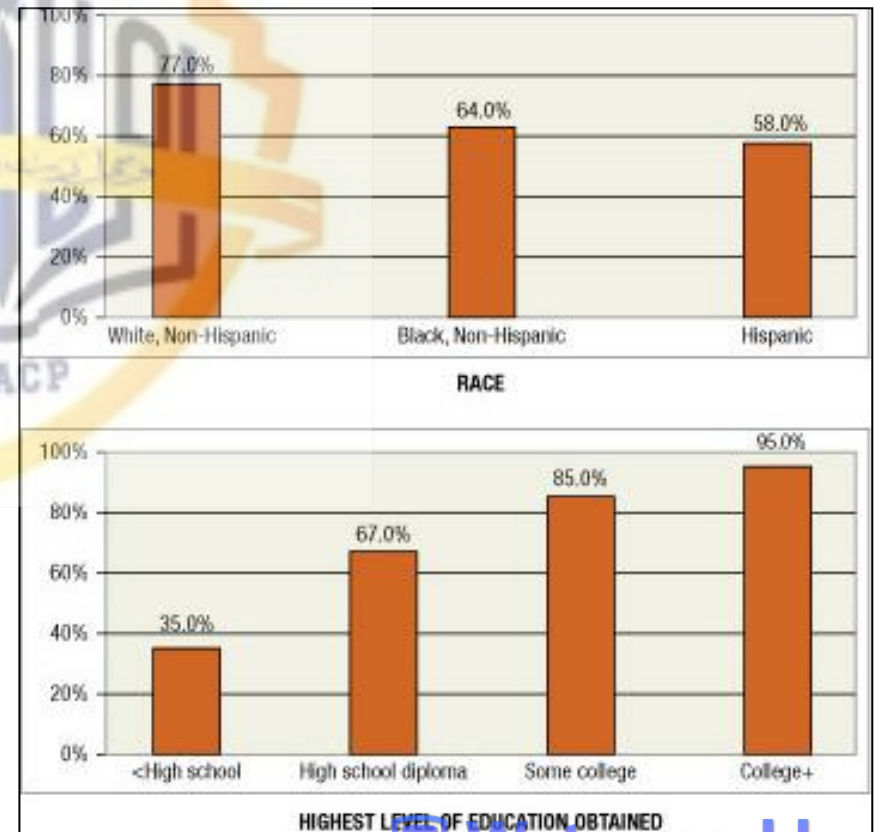
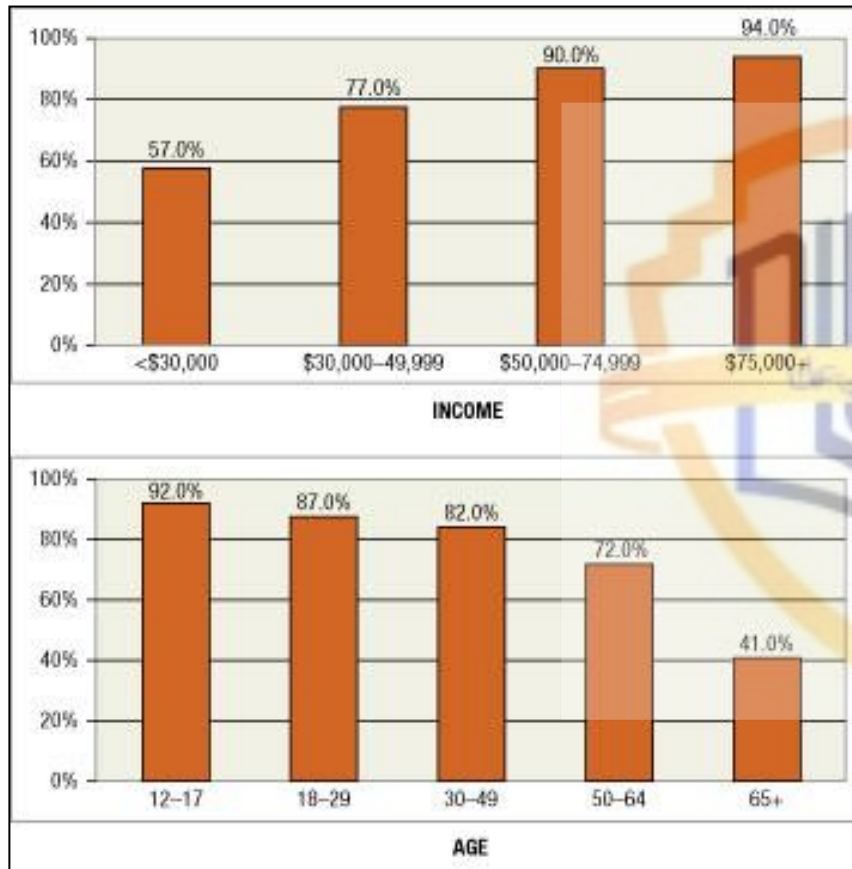


FIGURE 16-25
Key U.S. Internet use statistics. Shows the percent of individuals in each category who use the Internet.

Access to Technology

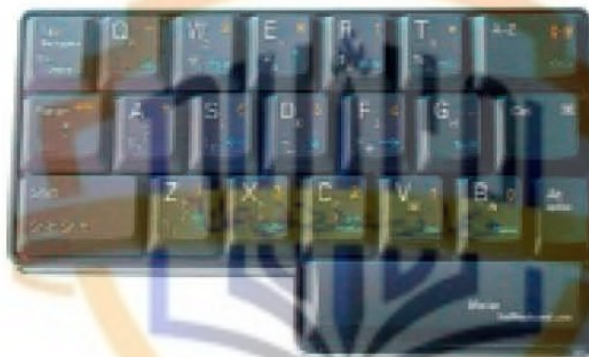
- Assistive technology: Hardware and software designed for use by individuals with physical disabilities
 - Assistive input devices include:
 - Braille keyboards, keyguards, one-handed keyboards
 - Voice input systems
 - Assistive pointing systems (switches, feet mice, head pointing systems, eye pointing systems, etc.)
 - Can also be used by general population as desired

Access to Technology



BRaille KEYBOARDS

The keys on this keyboard contain Braille overlays.



ONE-HANDED KEYBOARDS

Each key on this half keyboard contains two letters (one for the right half and one for the left) so all keys can be reached with one hand.



EYE TRACKING SYSTEMS

Cameras track the user's eye movements, which are used to select icons and other objects on the screen.

 **FIGURE 16-27**
Assistive input devices.

Access to Technology

- Assistive output devices include:
 - Screen readers
 - Braille displays
 - Braille printers
- Windows includes some accessibility features



Environmental Concerns

- Green computing: The use of computers in an environmentally friendly manner
- Energy consumption and heat are key concerns today
 - Energy Star program:
Developed to encourage the development of energy-saving devices
 - Eco-labels also used in other countries
 - IT energy consumption is key
 - Energy vampires



Environmental Concerns

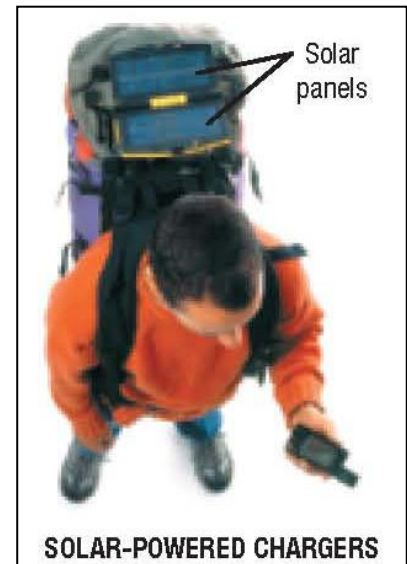
- Alternate power can be used with computers and other devices
 - Solar power
 - Hand chargers
 - Fuel cells



FIGURE 16-31
Alternate power.
Solar and hand power can be used to power mobile phones, portable digital media players, GPS devices, portable computers, and other devices.



HAND-POWERED CHARGERS



Environmental Concerns

- E-trash (e-waste): Electronic waste such as:
 - Disposable products (paper, disposable cameras, storage media, etc.)
 - Discarded hardware (old computers, phones, TVs, etc.)
 - Many elements found in hardware are dangerous
 - Arsenic, lead, mercury, cadmium
 - Much e-waste is shipped to other countries



FIGURE 16-32
E-trash.

The vast majority of the 40 million or so computers that become obsolete each year end up as e-trash in landfills.

Environmental Concerns

- Green computing methods
 - Recycling
 - Many devices, media, etc. can be recycled
 - Some states are implementing disposal fees for new hardware to ensure they are disposed of properly
 - Donate obsolete equipment to schools and other organizations
 - Data needs to be completely removed for security and privacy reasons
 - Reuse (recharged toner cartridges, new purpose for computers, etc.)

Quick Quiz

1. Which of the following is NOT an assistive input device?
 - a. Braille display
 - b. Head-pointing system
 - c. One-handed keyboard
2. True or False: E-trash is no longer a concern today since modern computers contain very few toxic materials.
3. A device designed to connect a portable computer to conventional hardware such as a keyboard, mouse, and printer is called a(n) _____.

Answers:

1) a; 2) False; 3) docking station

Summary

- Intellectual Property Rights
- Ethics
- Computers and Health
- Access to Technology
- Environmental Concerns
- Related Legislation

